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E-204 PDC SWITCHGEAR

1. SCOPE OF WORK

The work shall include the design, supply, factory testing, delivery to the Site, installation, testing and commissioning of all low voltage switchgear for distribution centers as part of the Work in accordance with Part 4, Subsection 1.2 Summary of Work.

The spare complete feeder shall be provided at least 20% for motor and outgoing feeder.

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of Part 4, Subsection 2.5, Codes and Standards, and the following specific standards.

2.1 National

PUIL : Peraturan Umum Instalasi Listrik - Indonesia (General Regulation of Electrical Installation - Indonesia).
SPLN : Standard Perusahaan Umum Listrik Negara.

2.2 International Electrotechnical Commission

IEC 60044 : Instrument transformers
IEC 60186 : Voltage transformers
IEC 60255 : Electrical relays
IEC 60439 : Low-voltage switchgear and controlgear assemblies
IEC 60947 : Low-voltage switchgear and controlgear

2.3 American National Standards Institute

ANSI C37.20 : Standard for Switchgear Assemblies Including Metal Enclosed Bus

3. OPERATING AND DESIGN CONDITIONS

3.1 Environment

All equipment shall be suitable for operation in a combined cycle power plant on the seacoast. Refer to Part 4, Subsections 1.1, 2.14 and 2.16 for meteorological, seismic and environmental conditions at the Site.

3.2 Equipment Ratings

- | | | |
|-----|--|------------------------------|
| (1) | Rated Insulation Voltage | : 600 V |
| (2) | Nominal Voltage | : 400 V |
| (3) | Maximum Voltage | : 440 V |
| (4) | Rated Frequency | : 50 Hz |
| (5) | Short-Circuit Performance Category | : P-2 |
| (6) | Rated Short-Circuit Current (interrupting) @ 0.15 power factor | |
| | a. With instantaneous trip | : 50 kA*) |
| | b. With delayed trip | : 50 kA*) |
| | c. Rated duration of short circuit | : 1 s |
| (7) | Rated making current | : 50 kA*)@ 0.15 power factor |
| (8) | Number of poles | : 3 |
| (9) | Nominal opening and closing mechanism voltage | : 220 V DC |

Note: *) Items to be determined by the Contractor and subject to approval by the Owner.

4. TECHNICAL REQUIREMENTS

4.1 General Requirements

PDC Switchgears shall be manufactured or assembled in Indonesia.

The switchgear shall consist of circuit breakers, buses and accessories, mounted on metal frames, complete with all electrical connections and completely enclosed within sheet-metal housings. The metal thickness shall be accordance with technical specification E-280.

All components shall be constructed from non-hygroscopic and non-flammable materials. The equipment shall be suitable for the loads specified without the need for forced cooling.

PDC Switchgears having tie breakers shall have interlocks for the tie and the two (2) main breakers to ensure that not more than two (2) of the three (3) breakers can be closed at the same time.

Common PDC Switchgear designed to feed from one of unit PDC Switchgear with interlocking system.

4.2 Structure and Housings

Each circuit breaker shall be installed in a separate compartment with a hinged door opening to the front of the housing. All doors shall have provisions for padlocking.

Enclosures of any outdoor switchgear shall be designed to ensure safe operation under all weather conditions. They shall be undercoated on the bottom to prevent corrosion. Space heaters with thermostatic control shall be installed to provide protection from the effects of moisture and low temperatures.

The switchgear shall have a degree of protection of at least IP 42 for indoor switchgear and IPW 54 for outdoor switchgear.

A ground bus shall be installed in each structure and bolted to each section of it. The ground bus dimensions shall in no case be less than 7 x 50 mm copper. A means shall be provided which will effectively ground the frames of all circuit breakers and other removable devices in both the operating and test positions.

4.3 Buses and Insulation

Main buses and conductors shall be copper and, when installed, shall be capable of carrying the maximum load continuously with a temperature rise

not exceeding the values specified. The bus system shall be able to carry rated short circuit current for one (1) second.

The phase designations of all terminals shall be clearly marked. All buses shall be arranged in the order R, S, T front to back, top to bottom, or left to right as viewed from the switchgear front (operating mechanism side).

4.4 Circuit Breakers

Circuit breakers shall be three-pole, single throw, air magnetic, electrically or manually operated as specified, drawout type, supported on a frame and designed to be rigidly connected to the housing in the operating position.

There shall be provision for easily moving the circuit breaker to a test position (by means of a hand-operated device at each breaker), when the main disconnecting devices are open, but with the control and operating circuits operable.

It shall also be possible to move the breaker to a fully disconnected position. It shall be possible to close the cubicle door with the circuit breaker inside in the disconnected position.

Interlocks shall be provided which shall prevent moving the circuit breaker from the fully connected, test, or fully disconnected position unless the breaker is open. The interlocks shall prevent the breaker from being closed between any of these positions.

Electrically operated closing and tripping mechanisms shall be designed for remote control at a nominal voltage of 220 V DC. Closing and tripping mechanisms shall operate satisfactorily over the following voltage range:

- | | | | |
|-----|------------------------------------|---|--------------|
| (1) | Closing mechanism operating range | : | 158 to 240 V |
| (2) | Tripping mechanism operating range | : | 125 to 240 V |

All circuit breakers shall be provided with stored energy operating mechanisms. It shall be impossible to close a breaker unless there is adequate energy stored to trip the breaker. The breaker springs of electrically operated breakers shall be charged after closing.

Provision shall be made for manual operation of circuit breakers from the front of the housing without opening the breaker compartment.

All circuit breakers shall be equipped with the followings:

- (1) A cell switch, which can be used to externally indicate when the breaker is in the fully connected position.
- (2) A lamp indicator or a target located so that the position of the breaker contacts, open or closed, is indicated without opening the door.
- (3) A “Stored energy” type of closing mechanism.
- (4) A bell alarm contact which closes when the breaker is tripped on overcurrent.

The breaker closing springs shall be automatically discharged when the breaker is withdrawn from its cubicle. A provision shall be made for manually charging the breaker springs of electrically operated circuit breakers for testing and maintenance.

(1) Main Circuit Breaker

Each main air circuit breaker shall be complete with:

- a. Twelve (12) circuit auxiliary switches operational in the “connected” and “test” position. Normally-open contacts shall be field convertible to normally-closed and vice versa.
- b. Three-pole, solid state, long time delay overcurrent trip device, adjustable between 80 to 120% of breaker trip rating with short time delay overcurrent selective trip attachment to provide selective tripping with bus tie circuit breakers and electrically and manually operated feeder circuit breakers.

(2) Electrically Operated Feeder Breaker

Each electrically operated feeder circuit breaker shall be complete with:

- a. Twelve (12) circuit auxiliary switches operational in the “connected” and “remote test” positions. Normally-open contacts shall be field convertible to normally-closed and vice versa.

- b. Three-pole, solid state, long time delay overcurrent trip device, adjustable between 80 to 120% of breaker trip rating to trip breaker between 7 and 25 second at five (5) times trip rating and instantaneous overcurrent trip device adjustable between 4 and 12 times breaker rating. Breakers shall include solid state ground detection devices.

(3) Manually Operated Feeder Circuit Breakers

Each manually operated circuit breaker shall be complete with:

- a. Three-pole, solid-state, long time delay overcurrent trip device, adjustable between 80 and 120% of breaker trip rating with short time delay overcurrent selective trip attachment to provide selective tripping with a motor control center or power distribution panel molded case circuit breaker.
- b. Shunt trip mechanism.

4.5 Protective Relays and Test Devices

Protective relays shall conform to the requirements of the applicable parts of the IEC 60255.

One (1) single element definite time ground overcurrent relay shall be provided for each of switchgear. The contacts shall be brought out to terminal blocks.

All switchgear with electrically operated circuit breakers shall be equipped with two (2) time undervoltage relays. Each relay shall have two (2) sets of contacts, one (1) wired to auxiliary relays and one (1) for alarms.

- (1) Each undervoltage relay shall have adjustable time delay and a pickup setting range from 0.6 to 0.8 per unit bus voltage.
- (2) The undervoltage auxiliary relays shall be connected to trip all electrically- operated breakers feeding motor loads on undervoltage.
- (3) If an AC static trip device is used, the necessary auxiliary transformers shall be provided.

4.6 Instruments

Each power distribution center shall be equipped with a voltmeter and a four (4) position meter switch, having positions of “Off”, “Phase R-S”, “Phase S-T”, and “Phase T-R”. The voltmeter shall be fed from the voltage transformers.

4.7 Instrument Transformers

Instrument transformers secondary windings shall be terminated at barrier type terminal blocks suitable for connection to external cables.

Current transformer secondaries shall be connected to shorting type terminal blocks. Temporary jumpers shall be provided at the terminal blocks to ground and to prevent an open circuit if the switchgear is energized.

Each set of switchgear shall be equipped with a set of voltage transformers for relaying and metering.

The primary and secondary of voltage transformers shall be fused except when an open-delta connection is required. The secondary of an open-delta configuration shall have the common phase grounded and the other phases fused. All instrument transformers shall be grounded directly to the ground bus, not to the housing.

4.8 Control Devices and Wiring

Terminal blocks and internal wiring shall meet the applicable requirements of Technical Specifications E-001, E-230 and I-002.

Each switchgear shall include the followings:

- (1) One (1) two-pole, double-throw, 60 ampere, 250 Volt DC knife switch for incoming control power. The switch shall have positions for connecting the switchgear to either of two (2) alternate feeds and a center off position.
- (2) One (1) DC control power failure relay, with contacts brought out to terminal blocks for alarm.

- (3) Two (2) time delay undervoltage relays as called for in Clause 4.5.
- (4) One (1) single element definite time ground over current relay as called for in Clause 4.5.

4.9 Nameplates

Nameplates shall be installed to designate the purpose of all circuits, instruments, meters, relays and fuses.

4.10 Temperature Rises

The temperature rise shall be in accordance with IEC 60947 and IEC 60439.

4.11 Tools

The following tools shall be provided for each switchgear room or isolated switchgear:

- (1) One (1) air circuit breaker removal device.
- (2) One (1) test cabinet with control switch, indicating lights and control cable connecting device.
- (3) One (1) manual operating device per power distribution center.
- (4) One (1) portable test apparatus for the circuit breaker static trip device..

4.12 Painting

The equipment painting shall be in accordance with Technical Specification E-001. The color of the finish coat shall be specified later.

5. PERFORMANCE, TESTING AND GUARANTEES

5.1 Type Tests

Type tests shall be performed on equipment identical to the types supplied under this Contract. The switchgear shall pass the type tests specified by IEC 60947.

5.2 Routine Tests

Routine tests shall be in accordance with IEC 60947.

6. QUALITY ASSURANCE/QUALITY CONTROL

Quality assurance/quality control shall conform to the requirements of Part 4, Subsection 2.7. shall include, but not be limited to, the followings:

- (1) Design Criteria and Engineering Calculations
- (2) Continuity and Wiring Tests
- (3) Short Circuit Tests
- (4) Insulation and High Voltage Tests
- (5) Packaging and Shipping
- (6) Welding

The documentation as specified shall include, but not be limited to, the followings:

- (1) Material Test Reports
- (2) Inspection Reports
- (3) Shop Test Reports

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